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# A paleomagnetic pole position of Late Jurassic/Early Cretaceous dolerites from Hinlopenstretet, Svalbard, and its tectonic implications

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Paleomagnetic examination of Late Jurassic/Early Cretaceous dolerites, Hinlopenstretet, reveal a mean paleofield direction ($D = 181°$, $I = -71°$, $\alpha_{95} = 4.2°$, $k = 117.9$, 17 sites, 96 samples) which is argued to represent a deuteric magnetization formed at the time of cooling. The resultant paleopole (66 ° N, 200 ° E, $dp = 7°$, $dm = 8°$) accords with paleomagnetic poles of similar age for North America after taking into account post-Jurassic continental separation. The result from the present study is interpreted to represent a Late Jurassic/Early Cretaceous paleopole for Eurasia. The similarity of the paleopoles from cratonic North America and Eurasia in its pre-drift configuration at about 145 Ma, makes it reasonable to conclude that relative displacement due to extension between Eurasia and North America prior to the onset of sea-floor spreading, must have taken place after 145 Ma and before the start of sea-floor creation in the Cretaceous.

## 1. Introduction

The Svalbard Archipelago lies on the western edge of the continental Barents shelf between Norway and the North Pole and belongs to the Eurasian plate (Fig. 1). As yet there are few reliable paleomagnetic pole positions from European Jurassic rocks and the available data are highly scattered [1–4]. Although the apparent polar wander path (APWP) for cratonic Eurasia is relatively well-defined during the Late Carboniferous–Early Triassic interval, it is poorly determined over much of the Mesozoic era.

In a study of Jurassic paleomagnetic poles from cratonic North America, May and Butler [5] consider eight results to be reliable Jurassic reference poles. These poles form a consistent chronological progression defining two arcuate tracks of APW in the time span 203–145 Ma. Combined with reliable Triassic and Cretaceous reference poles, the resulting path is significantly different from previous APW compilations. Paleomagnetic Euler pole (PEP) analysis [6] reveals two changes in the direction of APW: one in the Late Triassic to early Jurassic (J1 cusp) and one in the Late Jurassic (J2 cusp) [5]. May and Butler suggest that the J1 cusp represents the change in North American absolute plate motion associated with rifting of the central Atlantic, while the J2 cusp correlates with plate reorganization at marine magnetic anomaly M21 time.

The small number of Jurassic paleomagnetic poles from Europe combined with their large scatter, makes a chronological analysis difficult. Still one of the main aims in paleomagnetic research of Mesozoic rocks in Europe must be to increase the number of reliable paleopole determinations. This account reports on the paleomagnetism of selected dolerites from eastern Svalbard.

## 2. Geology

The geology of the Svalbard Archipelago is described in several papers [7–12]. The basement rocks of Spitsbergen, the largest island in the Svalbard archipelago, consist of Proterozoic to Paleozoic, Hecla Hoek successions which were metamorphosed and strongly deformed during the Caledonian orogeny. At Spitsbergen a large complex graben, bounded by north–south faults, is filled with Devonian red beds. To the south the red beds are overlain by a thick succession of Upper Paleozoic, Mesozoic and Tertiary sedi-

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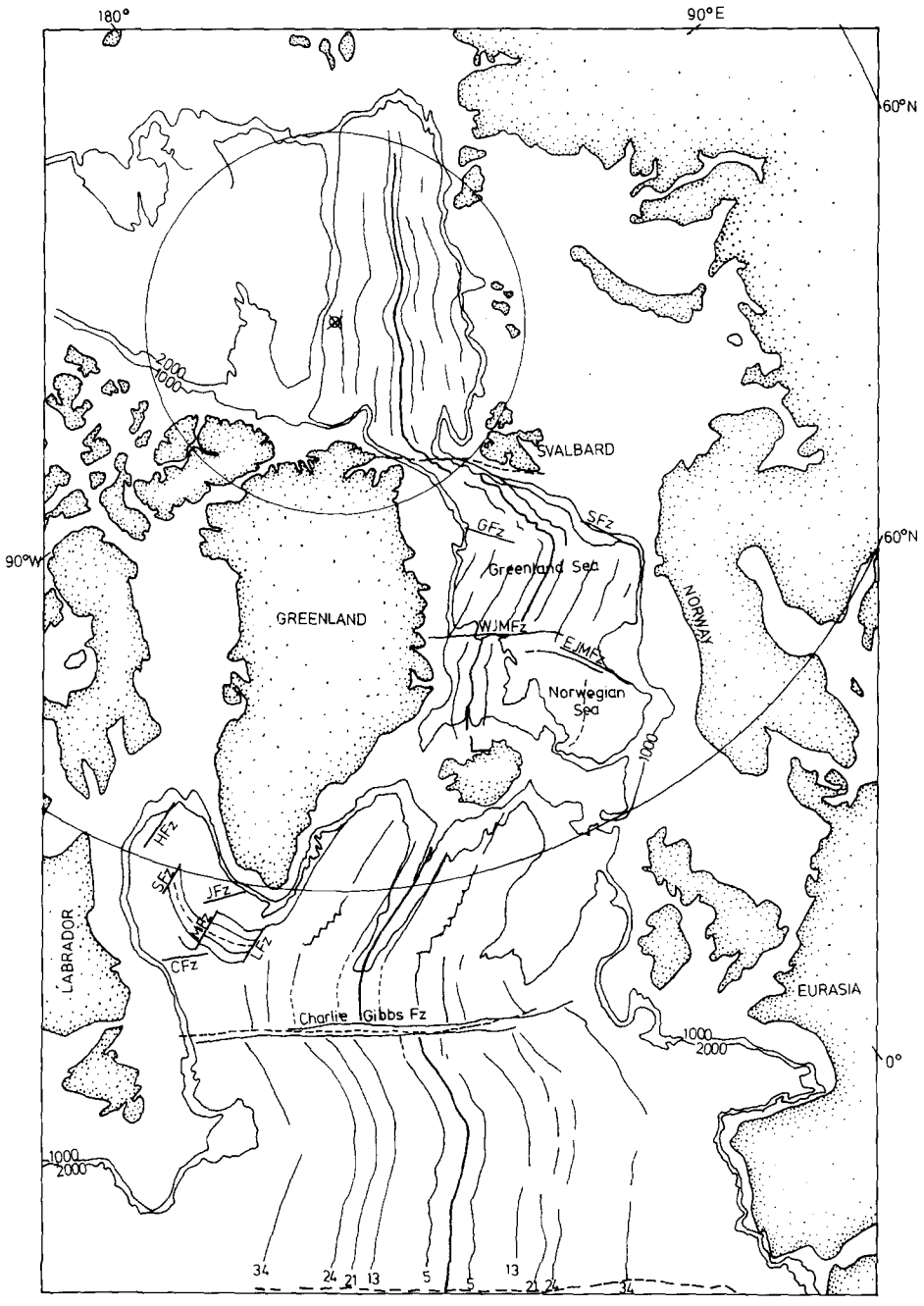



Fig. 1. Sketch map indicating major structural features in the North Atlantic, Norwegian–Greenland Sea and the Arctic Ocean. Fracture zones and magnetic lineations are shown by thick and thin lines. Also shown are 1000 m and 2000 m isobaths. (Simplified from [14].)

ments covering most of the southern half of the island. A Tertiary orogenic belt is exposed on the middle to south of the west coast.

Opening of the central Atlantic Ocean during the past 200 Ma separated North America and Africa. The beginning of sea-floor spreading (SFS) in the North Atlantic in the Early Cretaceous caused the separation of North America and

Eurasia. The plate kinematics of the entire central Atlantic and North Atlantic and the evolutionary history of the oceans have been described in two recent papers [13,14]. The major structural elements in the Norwegian–Greenland Sea and the Arctic Ocean are shown in Fig. 1. The crust between Norway, western Svalbard and Greenland was subjected to phases of tensional tectonics during the late Paleozoic/early Mesozoic without complete crustal break-up. Several basins were downfaulted during Early Cretaceous time along the North Atlantic rift system from Rockall Trough to southwest Barent Sea. Crustal stretching and thinning brought this deep central part of the rift system close to breakup by Early Cretaceous times. No major fault-activity is observed within Spitsbergen during this time interval, but intrusive and extrusive igneous activity is widespread.

Using the SFS magnetic anomalies and the fracture zones in the North Atlantic, Labrador Sea, Norwegian–Greenland Sea and the Eurasia Basin, Srivastava and Tapscott [14] have described the plate kinematics of these regions. Their study considers for the first time the entire boundary between the Eurasian and North American plates in determining the poles of rotation for the whole North Atlantic. The evolutionary history of the North Atlantic north of Azores began in early Cretaceous with the separation of Iberia from Newfoundland. Prior to anomaly 34 time SFS extended into the Rockall Trough and continued into the Labrador Sea in Late Santonian time.

With a change in direction of motion between Greenland and North America and between Greenland and Eurasia, just after anomaly 25 time, SFS started in the Norwegian–Greenland Sea and in the Eurasia Basin. Spreading in the Labrador Sea ceased prior to anomaly 13 time, which resulted in the change of direction of motion in the Norwegian–Greenland Sea. SFS in the Eurasia Basin has been along the Nansen Ridge since the beginning of spreading. The motion between Svalbard and Greenland was strike-slip until anomaly 13 time when it changed to spreading [14].

Igneous activity in the form of tholeitic dolerite sills, dikes and basaltic lava flows, is a characteristic feature of the platform evolution in Svalbard during the Mesozoic. Dolerite sills intruded primarily Carboniferous to Triassic sediments while dolerite dikes intrude in Precambrian to Late Jurassic rocks [11,15]. This magmatic activity is likely to be related to minor rifting events during the Jurassic. A geological map of the Hinlopenstretet area with the sampling localities is shown in Fig. 2.

## 3. Age determinations

Isotopic age determinations of four samples from Isfjorden and one from Hinlopenstretet by Gayer et al. [16] yielded ages ranging from $149 \pm 17$ Ma to $110 \pm 10$ Ma. Due to extensive hydrothermal alterations, these whole rock K/Ar ages are interpreted as minimum ages. The spread in ages (149–110 Ma) of the dolerites mainly from Isfjorden, can be shown to correlate with variation in hydrothermal alteration. It is thus likely that the "true" age is close to the oldest radiometric age, i.e. $149 \pm 17$ Ma. Parker [17,18] provides stratigraphic evidence for a post-Lower Volgian and pre-Valanginian (Late Jurassic to Early Cretaceous) age for the dolerites based on samples from Isfjorden and Agardbukta along the east coast of Spitsbergen. 42 K/Ar ages are reported by Burov et al. [19] from dolerites mainly from eastern Svalbard, they conclude that most intrusions were emplaced during two phases around $144 \pm 5$ Ma and $105 \pm 5$ Ma respectively. Undifferentiated dolerites were intruded during the first phase while in central Spitsbergen slightly differentiated intrusions were emplaced mainly during the second phase. Samples belonging to the 105 Ma group were altered, thus the ages should be regarded as minimum ages. Based on the reported radiometric and stratigraphic results it seems reasonable to suggest that the Hinlopenstretet dolerites are Late Jurassic/Early Cretaceous (144 Ma) in age. The timescale is from Harland et al. [20].

## 4. Petrology

The Hinlopenstretet intrusives are mainly grey medium- to fine-grained quartz dolerite with plagioclase laths which stand out against a dark grey background of pyroxenes and iron ores [16]. Irregular areas of serpentine enclosed in the pyroxenes reflect the former presence of olivine. Towards the marginal contact of sills and dikes, the minerals become progressively more finer

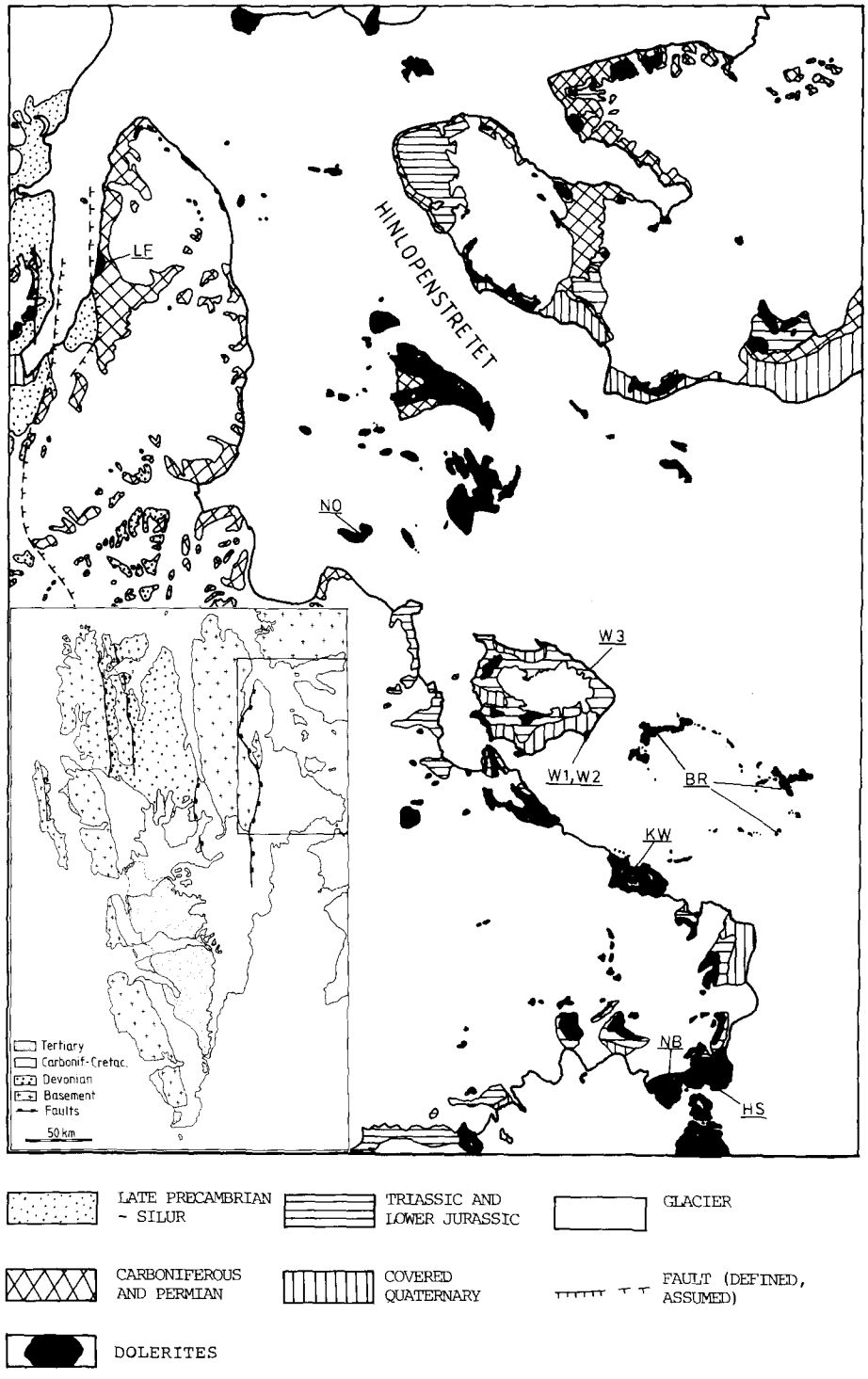



Fig. 2. Geological map of the Hinlopenstretet area with the sampled sites [28,29]. Included is a geological sketch map of Svalbard.

grained. The rocks are sparsely porphyritic and titanoferrous opaques are dispersed throughout the rock. The absence of olivine along the margin and the fine-grained texture is probably due to rapid cooling. The petrology and geochemistry of the Hinlopenstretet dolerites have been reported

TABLE 1

Geographic coordinates for the localities, and number of paleomagnetic samples/specimens

| Area | | | Number of samples | Number of specimens |
|---|---|---|---|---|
| Lomfjorden II | (79.5° N, 18.0° E) | LF | 15 | 23 |
| Wilhelmsøya I | (79.0° N, 21.0° E) | W1 | 15 | 19 |
| Wilhelmsøya II | | W2 | 7 | 17 |
| Wilhelmsøya III | | W3 | 11 | 26 |
| Kapp Weyprecht | | KW | 15 | 28 |
| Bastian- and Rønbeckøyane | | BR | 19 | 28 |
| Heyleysundet | (78.5° N, 21.0° E) | HS | 8 | 28 |
| Newtonbukta | | NB | 6 | 8 |
| Total | | | 96 | 177 |

by Weigand and Testa [21]. They examined samples from Lomfjorden and Wilhelmsøya, and found the samples to be tholeiitic in composition.

Reflected light and X-ray diffraction studies on the dolerites indicate that the opaques occur as titanomagnetite-ilmenite intergrowths, partly altered to titanomaghemite [17]. Investigation of the Lomfjorden dolerite [22] showed that the majority of titanomagnetite grains have exsolution lamellae of ilmenite formed by high-temperature oxidation. The area between the ilmenite lamellae consists of assemblages of Ti-poor titanomagnetite, relicts of the primary Ti-rich titanomagnetite or its low temperature oxidation product, cation deficient spinel (cds) phase, and heterogeneous black and grey spinels.

## 5. Sampling and laboratory results

Paleomagnetic samples were collected from four dolerite sills on the mainland, Spitsbergen (Lomfjorden II, Kapp Weyprecht, Heyleysundet and Newtonbukta), and two dolerite sills from Wilhelmsøya (Wilhelmsøya I, II and III). In addition, three dolerites were sampled from the Bastian- and Rønbeckøyane, Hinlopenstretet (Table 1, Fig 2.) Three to five handsamples were collected from each site, and 1–5 cores were subsequently drilled from each handsample.

The stability of the natural remanent magnetization (NRM) was tested by means of stepwise demagnetization, either alternating field (AF), or thermal demagnetization. Characteristic remanence components were calculated by a least square analysis method [23].

Thermomagnetic curves were obtained by using a horizontal Curie balance. Curie-temperatures range between 530° and 580°C (Fig. 3) and indicate Ti-poor titanomagnetite as the bulk magnetic phase in accordance with the results from the ore microscopy. Almost reversible heating/cooling curves indicate minor chemical changes during heating and that there is little titanomaghemite.

Specimens from almost all localities showed scattered dual-polarity directions of the NRM.

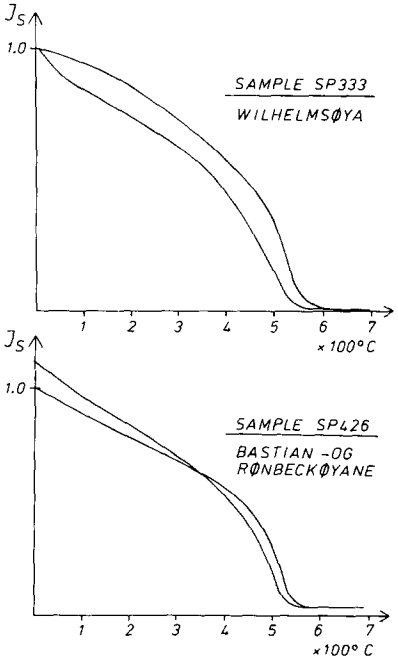



Fig. 3. Typical thermomagnetic curves obtained in air, indicative of single, high $T_c$ magnetic phases.

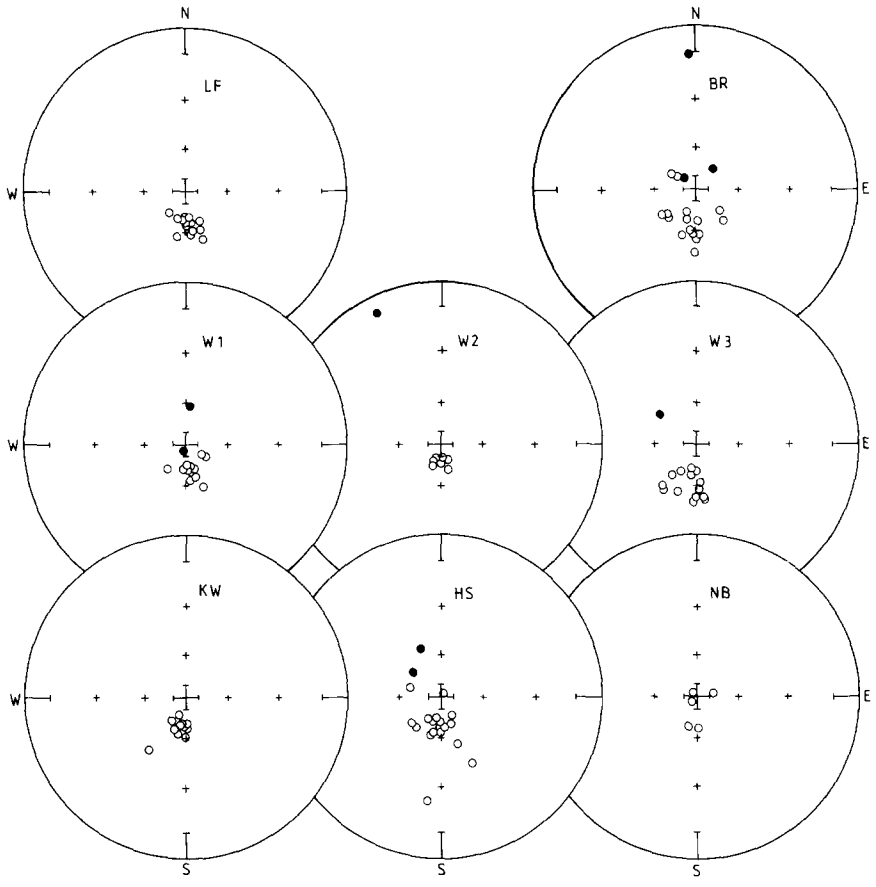



Fig. 4. Distributions of characteristic remanent directions from the Hinlopenstretet dolerites. Open (closed) symbols in the stereograms are up (down) pointing directions. Abbreviations: site regions as in Table 1.

NRM-intensities range between 6 and 5560 mA/m. AF and thermal demagnetization revealed pronounced differences in stability. Normal polarity directions were demagnetized in fields of less than 20 mT or temperatures less than 200 °C, whereas reversed polarity magnetizations are all associated with high blocking temperatures and coercivities.

All specimens from the upper Lomfjorden sill (LF) were AF demagnetizated revealing a single-component magnetization (Fig. 4, Table 2). From the island of Wilhelmsøya three profiles (W1, W2 and W3) were collected from two dolerites. A fairly well clustered reverse polarity group was identified from AF and thermal demagnetization (Table 2; Fig. 4).

AF demagnetization of the Kapp Weyprecht dolerite (KW) revealed single component stable paleomagnetic directions of reversed polarity (Table 2; Fig. 4).

Paleomagnetic results from Bastian- and Rønbeckøyane (BR) are mainly based on thermal demagnetization (Table 2; Fig. 4). Three specimens carried normal polarity directions, whereas the remainder are of reversed polarity.

The Heyleysundet dolerite (HS) shows mainly one group of stable reversed polarity directions (Table 2; Fig. 4).

Sample-directions from a single dolerite in Newtonbukta (NB) are more scattered than the others (Fig. 4; Table 2).

Demagnetization results from five specimens from Lomfjorden (SP255-E0), Wilhelmsøya (SP401-A2) and the Bastian- og Rønbeckøyane (SP413-B3, SP416-A2 and SP434-A1) are shown in Fig. 5 and Fig 6. Single polarity reversed NRM

TABLE 2

Paleomagnetic data for the Hinlopenstretet dolerites

| Area | $D$ (°) | $I$ (°) | $N$ | $\alpha_{95}$ (°) | $k$ |
|---|---|---|---|---|---|
| 1. Lomfjorden II | 172 | −67 | 18 | 3.2 | 104 |
| 2. Wilhelmsøya I | 168 | −72 | 16 | 3.8 | 87 |
| 3. Wilhelmsøya II | 179 | −78 | 8 | 2.9 | 292 |
| 4. Wilhelmsøya III | 189 | −61 | 17 | 4.7 | 53 |
| 5. Kapp Weyprecht | 191 | −69 | 20 | 3.0 | 112 |
| 6. Bastian- og Rønbeckøyane | 194 | −68 | 17 | 7.5 | 21 |
| 7. Heyleysundet | 187 | −66 | 18 | 7.0 | 22 |
| 8. Newtonbukta | 175 | −82 | 5 | 10.8 | 33 |
| 9. Lomfjorden I [22] | 171 | −68 | 38 | 3.0 | 96 |
| Overall mean Hinlopenstretet dolerites | 181 | −71 | 9 | 4.2 | 117.9 |
| VGP | 66 ° N, 200 ° E, ($dp = 7$, $dm = 8$) | | | | |

$D$ = declination, $I$ = inclination, $N$ = number of specimens, $\alpha_{95}$ and $k$ are calculated according to Fisher [27].
The mean value calculated from each area is based on characteristic magnetization components from specimens.
VGP: N = North latitude, E = East longitude, $dp/dm$ = semi-axis of the oval of 95% confidence about the mean pole.

directions associated with high $T_b$ dominate except in specimens SP401-A2 and SP434-A1 which reveal major unblocking below 400 ° C. Blocking temperatures normally varied between 400 ° and 560 ° C. This result compares well with the results from the thermomagnetic measurements which revealed Curie-temperatures between 530 ° and 580 ° C indicating Ti-poor titanomagnetite as the main magnetic mineral. The existence of normal polarity components is seen in specimen SP434-A1 in Fig. 6. These components are regarded as VRM magnetization representing the present field, they are removed at low temperatures or low AF fields. It was not possible to calculate any mean direction from these low stability components due to their large scatter.

Table 2 presents the overall paleomagnetic results for the Hinlopenstretet dolerites, and the site mean characteristic directions based on specimens are plotted in Fig. 7. It is assumed that the Hinlopenstretet dolerites were intruded at 144 Ma. Since the sediments are flat-lying no structural correction is necessary.

Since cation deficient spinel (cds) mineral phases are present in the rocks and only one single stable magnetization component is observed, a

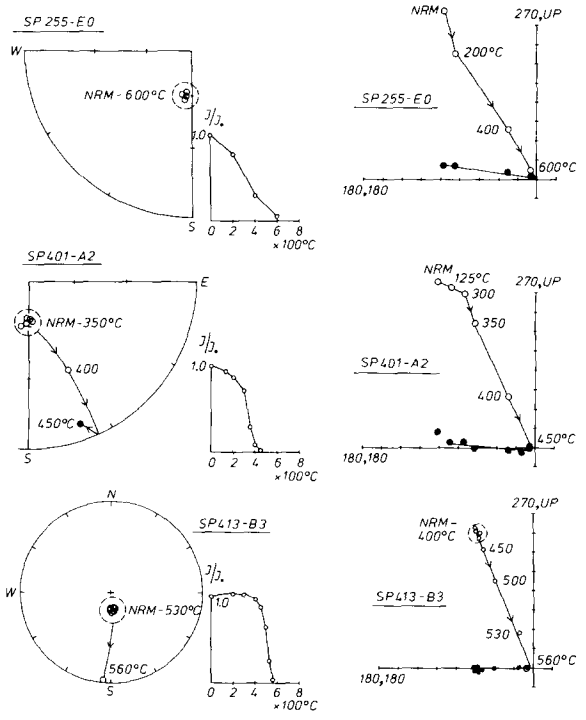



Fig. 5. Stepwise thermal demagnetization results from representative specimens from the Hinlopenstretet dolerites. LF: SP255-E0, W2: SP401-A2, BR: SP413-B3. Left: Directions in stereographic projections. Closed (open) symbols: lower/upper hemisphere. Middle: Intensity decay curves. Right: Orthogonal vector projections. Open circles are endpoint projections of demagnetization vectors on a vertical plane; solid circles on a horizontal plane.

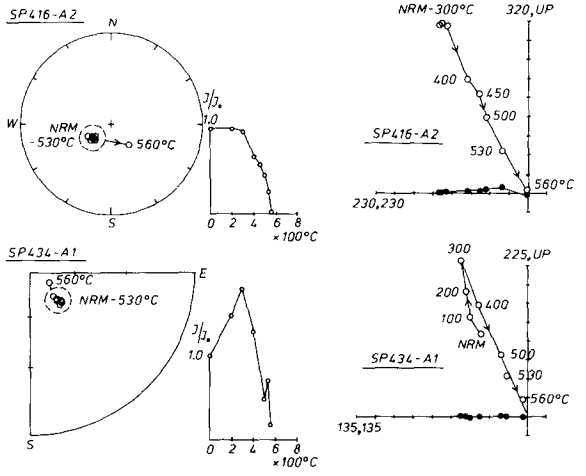



Fig. 6. Stepwise thermal demagnetization results from representative specimens from the Hinlopenstretet dolerites. BR: SP416-A2 and SP434-A1. Symbols as in Fig. 5.

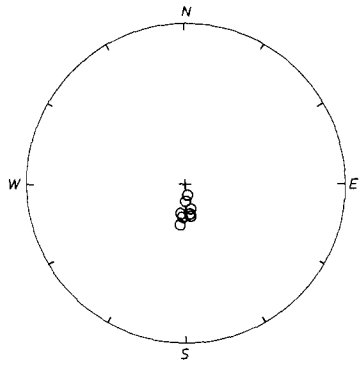



Fig. 7. Mean directions from the different areas.

deuteric origin of the cds phase is proposed. Both thermoremanent magnetization (TRM) acquired during cooling of the high-temperature oxidized titanomagnetites and a thermo-chemical remanence (TCRM) formed during the formation of the cds-phases are therefore likely to contribute to the observed magnetization.

The cooling time and/or emplacement rate may be too short to completely cancel the secular variation in the magnetization for each individual dolerite. It is, however, reasonable to assume that the secular variation is averaged in the overall mean value ($D = 181°$, $I = -71°$, $\alpha_{95} = 4.2°$ and $k = 117.9$) based on all the investigated dolerites. The results from the lower Lomfjorden dolerite from Briseid and Halvorsen [22] are incorporated in the overall mean, yielding a paleopole for the Hinlopenstretet (HI) located at 66° N and 200° E, $dp = 6°$, $dm = 7°$ (see Table 2).

## 6. Discussion and conclusion

The APWPs from Eurasia and North America have shown poor correspondence after 200 Ma when taking into account continental separation. The differences between the two APWPs have been attributed to poor definition of both paths in Jurassic and Early Cretaceous times. APW paths need continuous revision when new paleomagnetic and tectonic data become available. Precise geochronological data is also of great value especially in the Jurassic period with a usually large amount of apparent polar wander. The North American Jurassic apparent polar wander path has been reinterpretated [5] and eight paleopoles are considered to be reliable Jurassic reference poles for cratonic North America. These poles, which form a consistent chronological progression, are plotted in Fig. 8. It is evident that the North American APWP in the Jurassic is better determined than that of western Eurasia (Fig. 8). It seems reasonable to attribute the discrepancy in the APWP from the two continents to poor definition of the Western Eurasian APW path. The Jurassic paleopoles from Western Eurasia are all from rocks of Early to Middle Jurassic age and differ in age from the dolerites studied in the present paper.

Paleomagnetic results from the Hinlopenstretet dolerites reveal one stable magnetization component of reverse polarity ($D = 181°$, $I = -71°$, $A_{95} = 4.2°$ and $K = 117.9$) based on 157 specimens out of a total number of 177. The 157 specimens represent 96 samples from 9 dolerites covering the whole Hinlopenstretet area (Fig. 2). The results from the Lomfjorden sill are incorporated in the mean Hinlopenstretet (HI) paleopole since this result is based on one dolerite sill from the same area as the other results which are incorporated in the overall mean direction.

Magnetomineralogy, ore microscopy and blocking temperature spectra all point to Ti-poor titanomagnetite formed by high-temperature oxidation during initial cooling as the main carrier of the magnetization. Radiometric and stratigraphic age determinations have placed the intrusions of the dolerites at the Jurassic/Cretaceous border (144 Ma on the timescale of Harland et al. [20]). The intrusion of the dolerites may have taken place during chron M16/M17 on the geomagnetic polarity time scale of Harland et al. [20].

If the APWP for Eurasia is rotated about the Euler Pole for the complete opening of the Atlantic, it will be congruent with the part of the North American APWP that predates the opening of the Atlantic. If there were rifting before the formation of the oldest magnetic anomaly, the APWP will not be superimposed when using an Euler Pole for this magnetic anomaly. It has generally been assumed that the plates surrounding the North Atlantic have behaved rigidly after the mid-Paleozoic Caledonian orogeny, though geological observation on some of the plates, or along their margins does suggest that a portion of them have undergone deformation [3]. West Spitsbergen is affected by deformation in the Tertiary [24]

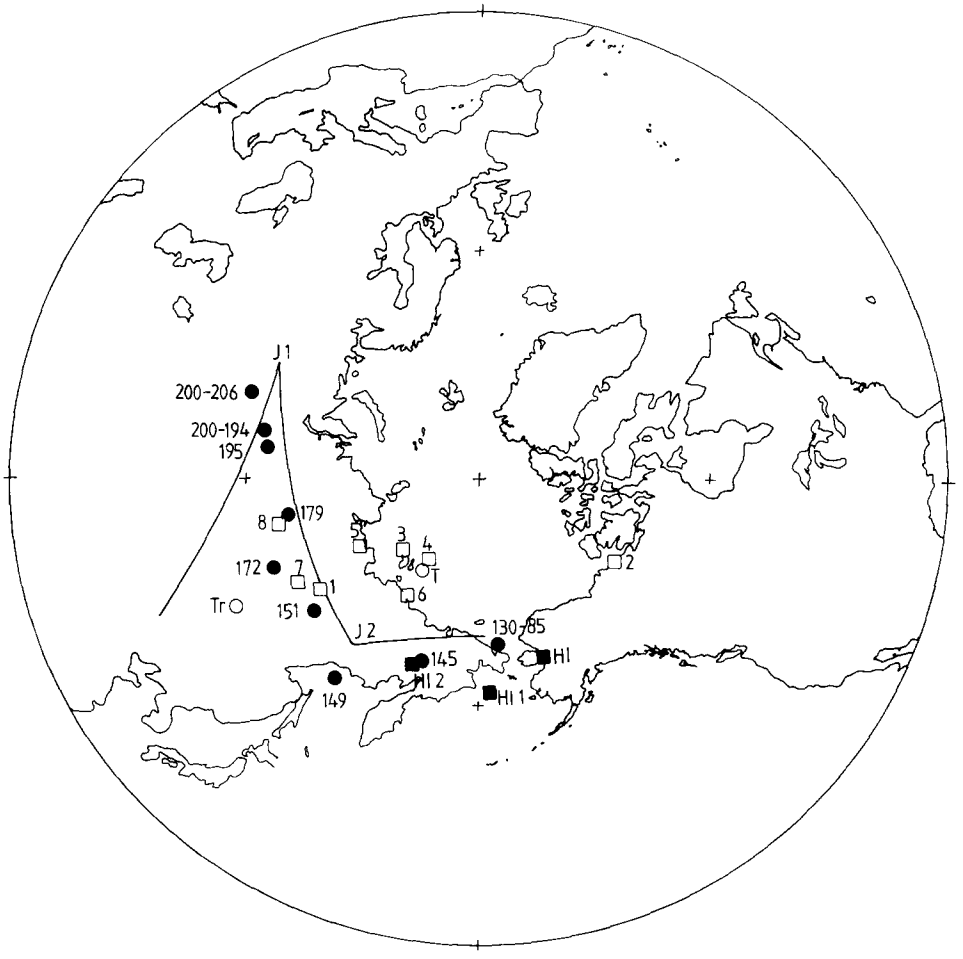



Fig. 8. Jurassic paleopoles from North America (filled circles) according to [5]. European data: *1* = [30], *2* = [31], *3* = [32], *4* = [33], *5* = [34], *6* = [35], *7* = [36] and *8* = [37]; *HI* = results from the Hinlopenstretet dolerites (this study). *HI1* and *HI2* show the position of the HI pole after rotating Eurasia according to the rotation data of [14] and [3] respectively.

while the eastern parts of the Svalbard archipelago with the Hinlopenstretet dolerites, only suffered minor near vertical fault movements along older lineaments.

The Hinlopenstretet paleopole is first rotated according to the rotation pole of Srivastava and Tapscott [14] based on sea-floor magnetic anomalies in the entire North Atlantic assuming that the opening followed a propagating rift model (rotation pole: 75°N, 152°E and rotation angle: −24°). Secondly the rotation data of Frei and Cox [3] is used which takes into account the relative displacement between Eurasia and North America prior to the formation of oceanic crust in the North Atlantic. The reconstruction is based on three independent data sets: SFS data, shelf lines and the APWP data set. They found that a rotation pole located at 88°N, 215°E with rotation angle −42° superimposed the two APWPs for the period 290–240 Ma and is reasonably consistent with the geology of the continental margins of the North Atlantic. Frei and Cox [3] suggest that the difference between their rotation pole and the one based on SFS data alone probably reflects relative motions of the two cratons in the time between Early Triassic and Late Cretaceous. Significant displacement probably took place in the North Atlantic area prior to the formation of oceanic crust [3].

The position of the Hinlopen paleopole after the first rotation is 63°N, 182°E and after the second 67°N, 161°E. The results are plotted in

Fig. 8. It is seen that the second rotation brings the paleopole in excellent agreement with the results from the Upper Morrison formation on the North American APWP and the ages are also in good agreement (145 Ma for the Upper Morrison Formation and 144 for Hinlopenstretet). From this result it is reasonable to conclude that relative displacement due to extension between Eurasia and North America prior to the onset of SFS must have taken place after 145 Ma.

This new paleopole makes it reasonable to conclude that the previous discrepancy in the APWP of Europe and North America around 145 Ma after taking into account post-Jurassic sea-floor spreading, is due to poor definition of the European APWP. The Hinlopenstretet paleopole (HI) is interpreted to represent a Late Jurassic/Early Cretaceous pole position for Eurasia. The paleopoles determined earlier (Fig. 8) reveal difference in latitude. Three poles (poles #1, 7 and 8) have latitudes comparable to the Hinlopenstretet paleopole, the longitudes are more westerly in accordance with the Lower to Middle Jurassic age for these poles. Four paleopoles (poles #3–6) cluster close to the position of Tertiary paleopoles for Eurasia (about 76° N, 161° E). These poles with magnetizations which are interpreted as primary are shown to be pre-folding. However, since the folding is believed to be Alpine, the magnetization age in these sediments can be Jurassic to Tertiary.

Based on sedimentologic data from Svalbard Steel and Worsley [12] found evidence for minor changes in latitudes for Spitsbergen in Late Jurassic/Early Cretaceous times. The small change in latitude compared to the longitudinal movement of the paleopoles from North America and Eurasia during the Late Mesozoic probably relates to the mainly circumpolar movement of the plates bordering the Arctic in that period [25]. The rifting, wrenching and spreading are mostly controlled by reactivation of zones of crustal weakness established prior to Permian times [26].

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